

CleanCSV Product Roadmap & Feature Plan

Transforming Automated CSV Data Cleaner into a Flatfile-Inspired Micro-SaaS

This roadmap outlines the prioritized phases to evolve the application from a single-file cleaner into a production-grade, Google-ranking web application with commercial SaaS potential.

Phase 1: Core Utility Upgrades (Immediate Priority)

Goal: Remove file format limitations and introduce initial Flatfile-style mapping.

1. Excel Support (.xlsx , .xls)

- Allow users to upload Excel workbooks alongside CSV files.
- Support sheet selection if an Excel workbook contains multiple tabs.
- Use `openpyxl` / `pandas` engine under the hood.

2. Smart Column Mapping (Flatfile-Lite)

- Display a visual mapping step after file upload.
- Detect source column headers (e.g. `User_Mail` , `First_Name`) and suggest standard target fields (`email` , `first_name`).
- Allow users to rename, reorder, or drop unwanted columns prior to cleaning.

3. Date & Time Standardization

- Automatically identify date columns formatted inconsistently (e.g., `12/05/2024` , `2024-05-12` , `May 12, 2024`).
- Standardize all dates into ISO 8601 format (`YYYY-MM-DD`).

4. Outlier Detection & Capping

- Use Interquartile Range (IQR) to flag impossible or extreme numeric values (e.g., negative salaries or age > 150).

- Allow user to choose: clip to boundaries, flag as null, or remove row.

Phase 2: Interactive In-Browser Data Grid & Validation

Goal: Provide instant, tactile control over dirty data.

1. Editable Table Cells

- Allow users to click directly into any cell to correct data on-screen before downloading.
- Implement lightweight virtual scrolling for fast rendering of large tables.

2. Cell-Level Error Highlighting

- Highlight invalid emails, broken URLs, and malformed phone numbers with visual error badges (red outline/tooltip).
- Provide a "Jump to Next Error" button to quickly navigate through rows requiring attention.

3. Multi-Format Export

- Export cleaned results to **CSV**, **Excel (.xlsx)**, **JSON**, or **Parquet**.

4. Visual Data Profiling & Audit Report

- Generate a downloadable summary (PDF or interactive drawer) showing:
 - Memory reduction percentage.
 - Missing data breakdown per column.
 - Value distributions (min, max, median, unique values).

Phase 3: AI-Powered Smart Cleaners (The "Pro" Differentiator)

Goal: Solve high-friction data problems that traditional regex cannot handle.

1. AI Categorical Fuzzy Fixer

- Unify misspelled categories into clean clusters (e.g., ["Dhk", "dhaka", "Dhaka City"] → "Dhaka").

2. AI Name & Address Dissector

- Split single-field full addresses into structured columns: Street , City , State , Postal Code , Country .

- Separate messy combined names ("Dr. John H. Watson Jr.") into Title, First Name, Last Name.

3. Custom Natural Language Prompts

- Allow users to type instructions like: "Extract domain name from all email addresses" or "Translate country names to English".

Phase 4: SEO, Architecture & Monetization (Launch to Market)

Goal: Capture organic Google traffic and establish revenue streams.

1. Programmatic SEO (pSEO) Landing Pages

- Create focused landing pages targeting high-intent long-tail keywords:
 - /clean-shopify-csv
 - /remove-csv-duplicates-online
 - /format-csv-dates
 - /convert-excel-to-clean-csv
- Include FAQ schema, how-to schema, and privacy assurances ("Processed in-memory. Zero server retention.").

2. Freemium Tier & Payment Gateway

- **Free Plan:** Files up to 5 MB, standard cleaning, single file.
- **Pro Plan (\$12/month):** Up to 100 MB files, batch folder processing, AI-powered cleaning, unlimited exports.
- Integrated with Stripe / LemonSqueezy.

3. Embeddable B2B JavaScript Widget (Flatfile Competitor)

- Provide an embed snippet (`<script src="../../../widget.js">`) that other software platforms can install to onboard their customers' CSVs directly.

Proposed Next Step

Select which feature to implement first:

- **Option A (Recommended):** Excel (.xlsx) upload and export support.
- **Option B:** Smart Column Mapping UI (match messy headers to target names).

- **Option C:** Date normalization & outlier filtering rules.